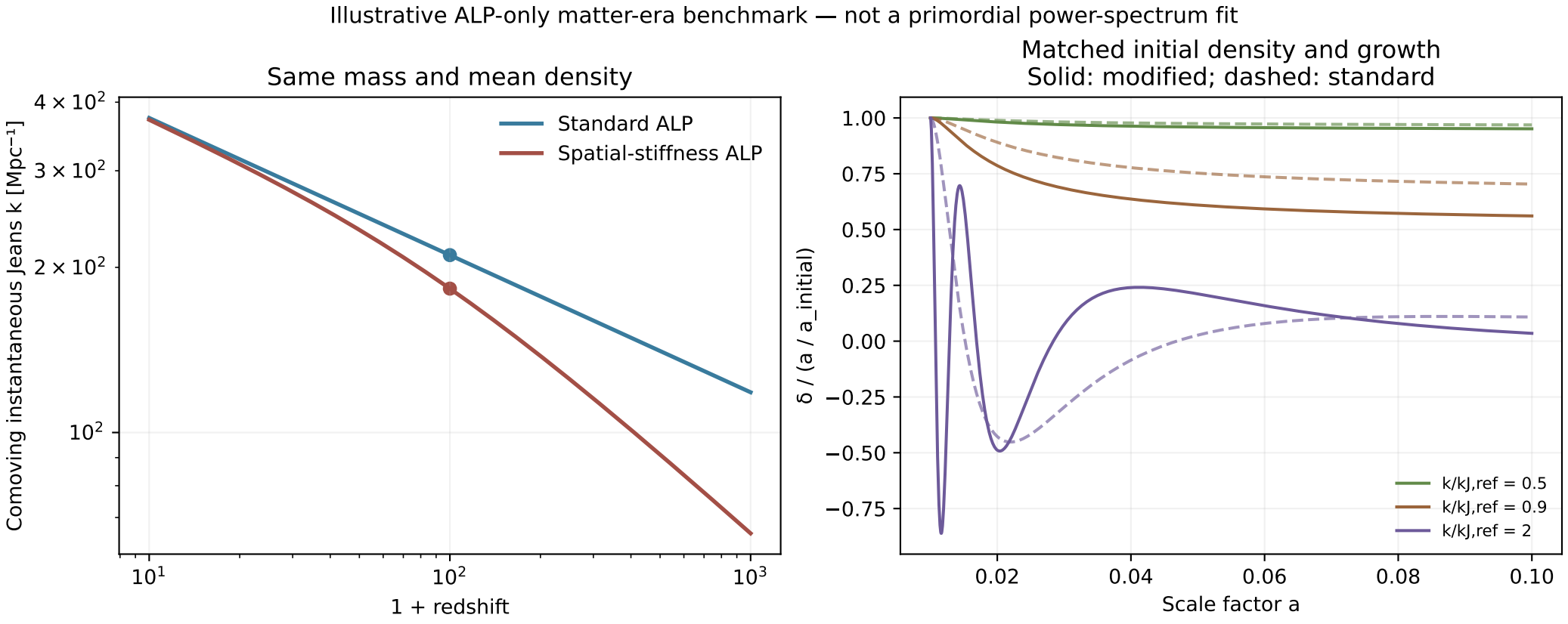
Illustrative ALP-only matter-era benchmark — not a primordial power-spectrum fit
Same mass and mean density
Standard ALP
Spatial-stiffness ALP
Comoving instantaneous Jeans k [Mpc⁻¹]
4 × 10²
3 × 10²
2 × 10²
10²
10¹
10²
10³
1 + redshift
Matched initial density and growth
Solid: modified; dashed: standard
δ / (a / a_initial)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
k/kJ,ref = 0.5
k/kJ,ref = 0.9
k/kJ,ref = 2
0.02
0.04
0.06
0.08
0.10
Scale factor a